

DECODER DESIGN MATTERS FOR ECG DELINEATION

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ABSTRACT

Electrocardiogram (ECG) delineation identifies the boundaries of P waves, QRS complexes, and T waves, providing structural annotations that can guide AI models in learning to interpret ECGs. However, training accurate delineation models requires manual annotations that are scarce and time-consuming to obtain. Recent work addresses this limitation through semi-supervised learning (SSL), but the design of the architecture, particularly the decoder, has received less attention. To this end, we propose R-U-Net, an ECG delineation model that pairs a ResNet-18 encoder with a U-Net decoder. On SemiSegECG, R-U-Net outperforms the strongest evaluated ResNet-18 + fully convolutional network (FCN) head baseline in each of the 16 in-domain settings by 3.3–13.0 mIoU and achieves 82.6 mIoU in the cross-domain setting, an improvement of 8.1 mIoU. Controlled ablations show that decoder design contributes more to performance gains than the evaluated SSL methods, motivating further exploration of architectures for ECG delineation. All code is open-source at github.com/ELM-Research/ECG-Delineation.

Index Terms— Electrocardiograms, ECG Delineation, Deep Learning, Semi-Supervised Learning

1. INTRODUCTION

Applying artificial intelligence (AI) to interpret ECGs is a step towards scalability and automation. While most recent works largely focus on tasks such as classification [1] and ECG-conditioned language generation [2], ECG delineation has been a growing area of interest [3].

ECG delineation is the task of partitioning the ECG in time into four categories: (1) background, (2) P wave, (3) QRS complex, and (4) T wave. Early automated ECG delineation methods relied on wavelet transforms and hand-designed rules [4]. More recent approaches formulate delineation as dense sample-level labeling using architectures such as CNN-LSTM models [5] and 1D U-Net models [6]. Waveform annotations provide explicit ECG structure and granular supervision for downstream tasks such as ECG-conditioned language generation [7]. However, training accurate delin-

R-U-Net

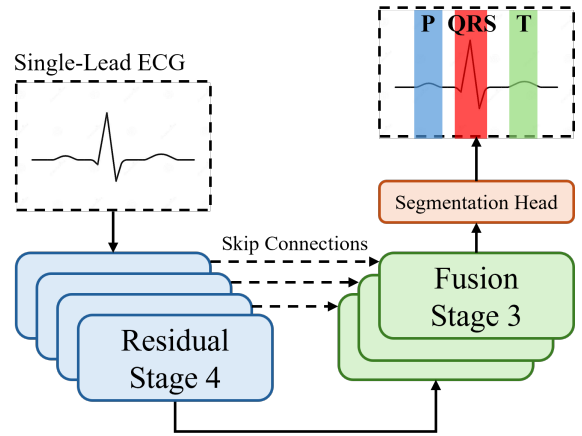


Fig. 1. A high-level architectural overview of R-U-Net.

ation models requires manual boundary annotations that are scarce and time-consuming to obtain.

To address this limitation, SemiSegECG [3] benchmarks semi-supervised learning (SSL) approaches for ECG delineation across six datasets. Its comparison primarily focuses on SSL strategies, pairing ResNet [8] and ViT [9] encoders with a lightweight fully convolutional network (FCN) head. Although U-Net architectures have previously been applied to ECG delineation, the contribution of decoder design within this benchmark remains unexplored.

In this study, we investigate decoder design for ECG delineation through R-U-Net, which pairs a ResNet-18 encoder with a U-Net decoder [10] in place of the FCN head. R-U-Net outperforms the evaluated baselines in all 16 in-domain settings and the cross-domain setting of SemiSegECG [3]. To isolate the contribution of decoder design, we conduct two ablation studies: (1) comparing decoder variants (Table 2) and (2) comparing SSL approaches (Table 3). In both studies, we find that the U-Net decoder contributes most to the performance gains. Altogether, these experiments highlight decoder design as a key factor in ECG delineation performance and motivate closer attention to architectural details.

^{*} Equal contribution

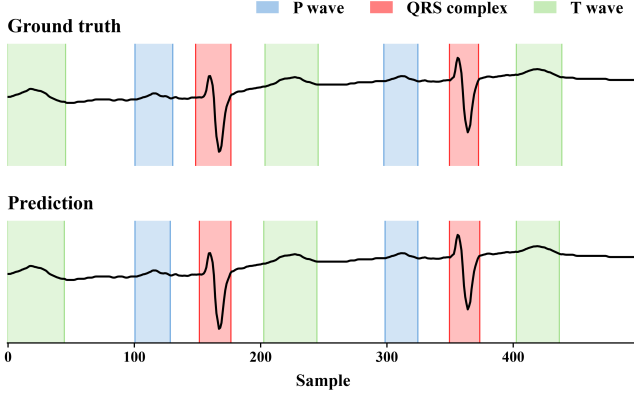


Fig. 2. Example of successful ECG delineation by R-U-Net compared with the ground truth, showing a 2-second segment from the middle of a 10-second ECG.

2. METHOD

2.1. Problem Formulation

We follow SemiSegECG and formulate ECG delineation as sample-wise classification. For a single-lead segment $x \in \mathbb{R}^T$, the target $y \in \{0, 1, 2, 3\}^T$ assigns each sample to background, P wave, QRS complex, or T wave, respectively. Given labeled segments $\mathcal{D}_l$ and unlabeled segments $\mathcal{D}_u$, we learn a model f_θ whose output includes a class-wise softmax:

$$f_\theta(x) \in [0, 1]^{4 \times T}, \quad \hat{y}_t = \arg \max_{c \in \{0, 1, 2, 3\}} [f_\theta(x)]_{c,t}. \quad (1)$$

The first and last samples of each contiguous run of a non-background class define its predicted onset and offset.

2.2. R-U-Net Architecture Details

R-U-Net combines a one-dimensional ResNet-18 encoder [8] with a U-Net-style decoder [10], both initialized from scratch. Figure 1 provides an overview of the architecture.

The encoder contains four residual stages, each comprising two basic residual blocks. Given an augmented input $\tilde{x}$, we retain the output of every stage:

$$H_{\text{enc}} = (H_{\text{enc}}^{(1)}, \dots, H_{\text{enc}}^{(4)}) = E_{\theta_{\text{enc}}}(\tilde{x}),$$

where $H_{\text{enc}}^{(s)} \in \mathbb{R}^{d_s \times T_s}$. The stage widths are (64, 128, 256, 512), with corresponding temporal lengths (625, 313, 157, 79) for $T = 2500$.

Starting from the deepest encoder representation, the decoder progressively upsamples the features and concatenates them with the corresponding encoder outputs through skip connections. Each fusion stage applies two kernel-size-3 convolutions, each followed by batch normalization and ReLU. The three stages produce 256, 128, and 64 channels, respectively.

The final decoder features are passed to a segmentation head. Dropout with probability 0.1 is followed by a pointwise convolution that produces $C = 4$ class logits. The head then linearly interpolates these logits to the original signal length and applies a class-wise softmax to obtain the predicted class probabilities:

$$f_\theta(\tilde{x}) = \text{softmax}_{\text{class}} (\mathcal{U}_T (D_{\theta_{\text{dec}}} (H_{\text{enc}}))),$$

where $D_{\theta_{\text{dec}}}$ comprises the decoder, dropout, and pointwise classifier, and $\mathcal{U}_T$ denotes linear interpolation to length T .

2.3. Boundary-Aware Mean Teacher

We use Mean Teacher [11] with a student f_θ and a teacher $f_{\bar{\theta}}$. The teacher is initialized from the student and updated after each optimization step as $\bar{\theta} \leftarrow 0.99\bar{\theta} + 0.01\theta$. It operates in evaluation mode without gradient updates. For each unlabeled segment, the teacher receives a weak view x^w obtained by random temporal resizing and padding or cropping. The student receives a strong view x^s with additional stochastic amplitude perturbations and powerline, white, or sinusoidal noise. The two views remain temporally aligned. Labeled segments and their masks undergo the same weak transformation.

For a labeled minibatch $\mathcal{B}_l$, we use sample-wise cross-entropy:

$$\mathcal{L}_{\text{sup}} = -\frac{1}{|\mathcal{B}_l|T} \sum_{(x,y) \in \mathcal{B}_l} \sum_{t=1}^T \log[f_\theta(x)]_{y_t,t}, \quad (2)$$

where (x, y) denotes an augmented segment and its aligned mask. For each unlabeled segment, write $q = f_{\bar{\theta}}(x^w)$ and $p = f_\theta(x^s)$. Both consistency terms use the soft-target cross-entropy $\ell_t = -\sum_{c=0}^3 q_{c,t} \log p_{c,t}$.

To emphasize potential waveform boundaries, we measure changes between adjacent teacher probability vectors:

$$\delta_t = \frac{1}{2} \sum_{c=0}^3 |q_{c,t} - q_{c,t-1}|, \quad t = 2, \dots, T, \quad (3)$$

with $\delta_1 = 0$. We spread these changes over a ± 4 -sample neighborhood and normalize over the full segment:

$$e_t = \max_{1 \leq u \leq T} \delta_u, \quad b_t = \frac{e_t}{\max(10^{-6}, \max_u e_u)}. \quad (4)$$

Let $a_t = \max_c q_{c,t}$ denote teacher confidence. Set $g = 1$ if its mean over the segment is at least 0.50, and $g = 0$ otherwise. The region and boundary weights are

$$w_t^r = g(1 - b_t) \mathbf{1}[a_t \geq 0.80], \quad w_t^b = gb_t(1 + a_t)/2. \quad (5)$$

The region term favors confident positions away from probability changes, including background. The boundary term

emphasizes these changes without the hard sample-wise confidence threshold.

For an unlabeled minibatch $\mathcal{B}_u$, we normalize the two terms independently:

$$\mathcal{L}_k = \frac{\sum_{x \in \mathcal{B}_u} \sum_{t=1}^T w_t^k \ell_t}{\max\left(1, \sum_{x \in \mathcal{B}_u} \sum_{t=1}^T w_t^k\right)}, \quad k \in \{r, b\}, \quad (6)$$

where the dependence of w_t^k and ℓ_t on x is implicit. The student minimizes

$$\mathcal{L} = \frac{1}{2} (\mathcal{L}_{\text{sup}} + \mathcal{L}_r + \mathcal{L}_b). \quad (7)$$

3. EXPERIMENTAL SETTINGS

3.1. Datasets

We follow the public in-domain and merged cross-domain protocols of SemiSegECG [3], using its supplied training, validation, and test splits. LUDB [12], QTDB [13], ISP [14], and Zhejiang [15] provide delineation labels. Each lead is treated as an independent input. Under the in-domain protocol (Table 1), labeled and unlabeled data come from the same dataset. Random subsets comprising 1/16, 1/8, 1/4, or 1/2 of the training set serve as labeled data, while the entire training set serves as unlabeled data. Under the merged cross-domain protocol (Figure 3), the four labeled datasets are combined while preserving their original splits, and PTB-XL [16] serves as an external unlabeled dataset. We evaluate on the merged in-domain test set to assess performance when labeled and unlabeled training data come from different sources. Following SemiSegECG preprocessing, waveforms are resampled to 250 Hz, producing $T = 2500$ samples, and processed with high-pass and low-pass filters at 0.67 and 40 Hz. Z-score normalization is applied to all model inputs.

3.2. Training and Evaluation

We train for 100 epochs using AdamW with a learning rate of 10^{-3} and weight decay of 0.05. The learning rate increases linearly during the first 10 epochs and subsequently follows a cosine schedule toward 10^{-4} . Each step uses 16 labeled and 16 unlabeled examples. We select the student checkpoint with the highest validation mean intersection-over-union (mIoU) and evaluate it on the test set. mIoU includes all four classes, including background. All mIoU results with standard deviations are from our experiments and are reported as the mean $\pm$ standard deviation across three random seeds.

4. RESULTS

4.1. In-Domain Evaluation

Table 1 presents the 16 in-domain evaluations on SemiSegECG. We compare R-U-Net trained using Boundary-aware MT

Table 1. In-domain test mIoU (%) under varying labeled-data ratios. Baseline results are from SemiSegECG [3] and use a ResNet-18 + FCN. R-U-Net results are reported as mean $\pm$ standard deviation over three seeds.

Method	ResNet-18			
	1/16	1/8	1/4	1/2
LUDB				
Scratch	67.3	71.3	72.9	74.1
MT	70.8	72.3	73.6	74.3
FixMatch	70.9	72.2	72.9	74.1
CPS	68.6	71.6	73.1	74.3
ReCo	71.5	72.5	73.1	73.9
ST++	69.2	71.6	73.7	74.5
R-U-Net	84.5$\pm$0.1	85.5$\pm$0.1	85.5$\pm$0.3	85.9$\pm$0.2
QTDB				
Scratch	47.5	56.2	60.5	64.9
MT	47.8	44.8	63.0	66.7
FixMatch	46.7	53.3	58.2	66.3
CPS	53.2	57.1	64.7	68.0
ReCo	53.4	53.4	58.7	64.5
ST++	52.9	57.8	62.5	68.0
R-U-Net	62.3$\pm$3.0	67.7$\pm$0.5	75.1$\pm$0.1	76.8$\pm$0.6
ISP				
Scratch	62.3	64.6	68.1	69.3
MT	64.2	65.9	68.0	69.1
FixMatch	63.1	64.4	67.9	68.9
CPS	64.4	66.1	68.5	69.4
ReCo	62.3	64.7	67.4	68.1
ST++	63.7	65.7	68.5	69.5
R-U-Net	76.7$\pm$0.2	77.9$\pm$0.2	79.3$\pm$0.2	79.9$\pm$0.1
Zhejiang				
Scratch	76.7	78.9	80.7	82.3
MT	79.2	80.1	81.5	82.9
FixMatch	79.0	80.4	81.2	82.7
CPS	77.6	79.5	81.3	82.7
ReCo	77.6	78.9	79.7	80.6
ST++	78.8	79.8	81.9	82.9
R-U-Net	83.7$\pm$0.1	84.8$\pm$0.1	85.4$\pm$0.0	86.2$\pm$0.1

with six baseline training methods using the ResNet-18 + FCN architecture, as reported in SemiSegECG [3]. R-U-Net achieves the highest mIoU across all four datasets and four labeled-data proportions, outperforming the strongest baseline in each setting by 3.3 to 13.0 mIoU. At the lowest labeled-data proportion (1/16), the improvements are 13.0, 8.9, 12.3, and 4.5 mIoU on LUDB, QTDB, ISP, and Zhejiang, respectively. These results demonstrate consistent improvements across datasets and levels of labeled-data availability, including settings with limited annotations.

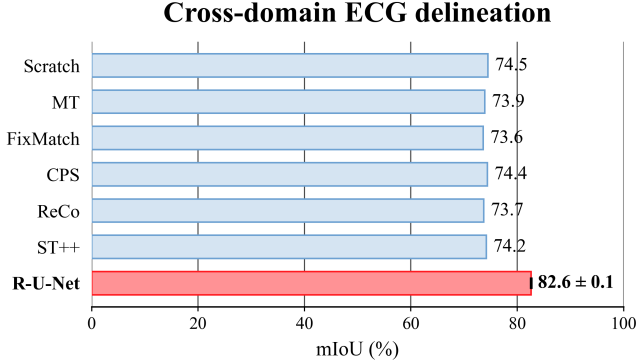


Fig. 3. Cross-domain test mIoU on the merged in-domain test set. Baselines use an FCN decoder [3].

Table 2. Ablation study of decoder variants on the LUDB dataset [12] using a labeled-data proportion of 1/16.

Decoder	mIoU
FCN	67.3
Wide FCN	67.2±0.3
U-Net (w/o skip connections)	80.0±0.1
U-Net	81.8±0.2

4.2. Cross-Domain Evaluation

Figure 3 compares R-U-Net trained using Boundary-aware MT with six baseline training methods using the ResNet-18 + FCN architecture in the cross-domain setting. R-U-Net achieves the highest mIoU of 82.6, exceeding the strongest baseline, Scratch (74.5 mIoU), by 8.1 mIoU. These results show that R-U-Net maintains its performance advantage when labeled and unlabeled training data come from different sources, supporting its effectiveness beyond the in-domain training setting.

4.3. Comparing Decoder Variants

We evaluate two intermediate decoder variants: (1) **Wide FCN**, which adds an additional convolutional layer and increases the channel widths to match the parameter count of our U-Net decoder ($\sim 1.03M$), and (2) **U-Net (w/o skip connections)**, which is the same U-Net decoder used in R-U-Net without skip connections. To isolate the effect of the decoder, we train Wide FCN, U-Net w/o skip connections, and U-Net with the scratch training method. Wide FCN achieves comparable performance to the original FCN (67.2 versus 67.3 mIoU). The U-Net decoder without skip connections achieves 80.0 mIoU, while adding skip connections improves mIoU by 1.8 points to 81.8. These results suggest that the gains primarily arise from the U-Net decoder, with skip connections providing a minor increase.

Table 3. Ablation study on applying different semi-supervised training methods to R-U-Net. We conduct experiments on LUDB 1/16.

Architecture	Method	mIoU
ResNet-18 + FCN	Scratch	67.3
	MT	70.8
	FixMatch	70.9
	CPS	68.6
	ReCo	71.5
	ST++	69.2
R-U-Net	Scratch	81.8±0.2
	MT	84.0±0.1
	FixMatch	83.5±0.2
	Boundary-aware MT (Ours)	84.5±0.1

4.4. Comparing SSL Approaches

Table 3 compares different training methods for R-U-Net with the ResNet-18 + FCN baselines on LUDB 1/16. Under scratch training, replacing the FCN decoder with the U-Net decoder increases mIoU from 67.3 to 81.8, a gain of 14.5 mIoU. Scratch R-U-Net also exceeds the strongest FCN baseline, ReCo (71.5 mIoU), by 10.3 mIoU. SSL provides smaller additional gains: FixMatch, standard MT, and boundary-aware MT achieve 83.5, 84.0, and 84.5 mIoU, respectively. Boundary-aware MT improves over scratch training by 2.7 mIoU and standard MT by only 0.5 mIoU. These results indicate that decoder design accounts for most of the improvement in this setting, with SSL providing modest additional gains.

5. CONCLUSION

In this paper, we introduce R-U-Net, which pairs a ResNet-18 encoder with a U-Net decoder for ECG delineation. Across the SemiSegECG benchmark, R-U-Net outperforms the evaluated ResNet-18 + FCN baselines in all 16 in-domain settings and the cross-domain setting, demonstrating consistent improvements across datasets and levels of labeled-data availability. Controlled ablations indicate that decoder design accounts for most of these gains. Increasing the FCN decoder’s parameter count provides no improvement, whereas the U-Net decoder substantially improves performance even without skip connections. Under scratch training, R-U-Net also exceeds the strongest evaluated FCN baseline by 10.3 mIoU points. Together, these findings highlight the importance of establishing strong architectural baselines when assessing SSL for ECG delineation, particularly under limited supervision. Future work could examine whether these decoder-level findings extend to other encoders and datasets, and develop SSL methods that further improve delineation when annotations are scarce.

6. COMPLIANCE WITH ETHICAL STANDARDS

This study retrospectively analyzed publicly available, de-identified ECG data from LUIDB [12], QTDB [13], ISP [14], Zhejiang [15], and PTB-XL [16]. No additional ethical approval was required for this secondary analysis.

7. ACKNOWLEDGMENTS

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